



Are loan sequences different from foreign sequences? A perception study with Japanese listeners on coronal obstruent – high front vowel sequences

Silke Hamann, Andrea Alićehajić

University of Amsterdam, The Netherlands
silke.hamann@uva.nl, andrea.alicehajic@gmail.com

Abstract

Native phonotactics influences speech perception, as numerous studies have shown. The present study tackles the question whether there is a difference in perceptual performance if the involved sequence occurs only in loanwords, compared to a sequence that does not occur at all in the native language. This was tested with the native Japanese sequences of palatal affricate plus /i/, compared to /ti/ (accepted only in loanwords) versus /zi/ (not accepted in Japanese) in an online AX discrimination task with 39 Japanese speakers (21-63 years old), who also had to answer three questions on their received English input. Participants performed significantly better at discriminating the accepted loan sequence /ti/, though discrimination of the foreign sequence /zi/ was also quite high (ranging from 40-100% correct). The results indicate that discriminability is only partly guided by native phonotactics. A potential role of amount of English input measured by self-report could not be attested.

Index Terms: Speech perception, Japanese, discrimination task, phonotactic restrictions, loanword adaptation

1. Introduction

The perception of an unfamiliar sequence in an unknown or new language is strongly influenced by our native phonotactic restrictions. This has been anecdotally reported already in 1931 [1], and experimentally supported by many later studies, for example, [2], [3], [4] and [5], showing that participants usually have problems distinguishing between structures that are illegal in their native language and similar structures with a repair strategy applied (such as introduction of an illusory vowel, or changes in a consonant or a vowel). The study by Dupoux et al. [2], e.g., found that native Japanese listeners performed poorly in discriminating words like [ebzo] from [ebuwo] (due to illegal structure */bz/ in Japanese), while native French listeners performed poorly in discriminating words differing in a vowel duration contrast only (due to the lack of a phonemic vowel length contrast in their language; */V:/).

At the same time, new phoneme sequences are regularly introduced into languages via loanwords due to extended contact between a phonotactically more restricted borrowing language and a less restrictive source language [6]. This is particularly obvious with the rising influence of English via social media and technology, bringing new words and phonotactics into languages with a stricter syllable structure. The question then arises how new sequences can be borrowed if they cannot be correctly perceived by the native listeners. A potential answer is that these borrowings are performed by bilingual speakers who have sufficient competence in both languages [e.g., 6, 7]. However, loan sequences are not only

used by bilingual speakers, hence the perception of all speakers of the borrowing language must have shifted over time to allow for formerly illegal sequences to enter the language, as argued by [8]. The present study looks into the question how this native perception changes, and more specifically, whether native speakers are more sensitive to phonotactic sequences that have been introduced already into their native language via loanwords than to sequences that are (not yet) nativized. This is done by comparing how well native speakers are able to discriminate a foreign language sequence that has been integrated into their language via loanwords from its closest native sequence, and whether this performance is similar to discriminating a sequence from the same foreign language that hasn't entered the native language yet.

The language of focus in this study is Japanese. Japanese is known for its restrictive phonotactics, though several previously impossible consonant-vowel sequences entered the language over the last centuries through loanwords from English, most likely due to increased contact with that language [9, 10, 11]. A well-known example for such a phonotactic restriction occurring only in loanwords is the sequence /ti/. Before 1890, this sequence was adapted into Japanese with palatalization and affrication of the plosive, as in the English word *team* borrowed as /tei:.mu/, following native phonotactic restrictions. From 1930 onwards, new loans from English with this sequence were adapted as /ti/, e.g. *party* /pa:.ti:/ [12, 13], introducing a new syllable to the Japanese language.

The sequence /zi/, on the other hand, has not yet been attested in Japanese pronunciation dictionaries or similar, with the exception of the careful articulation of the North American letter 'Z' [14], and no examples containing /zi/ can be found in recent loanword accounts (e.g., Appendix II of [15]). A previous study [16] showed that Japanese native listeners are better than chance at discriminating [dzi] – [zi] in non-words, with an average performance of 77 percent correct. [16] failed to compare this performance to established loan sequences to enable a conclusion on whether the found performance is average for all "foreign" phonotactic sequences (existing or non-existing in the language), or whether it is below the performance for sequences that occur in existing loanwords. [16] tried to explain potential variation in performance by the age of the participants, with the hypothesis that older speakers would perform worse than younger speakers as they are expected to have had less English input and hence less experience with the sequence [dzi]; however, the influence of age on performance was not significant in that study.

The present study compares how well Japanese listeners discriminate [dzi] – [zi] with how well they perform on [tei] – [ti], with the expectation that they are worse at the former as it contains an illicit sequence in Japanese.

It must be noted that for the Japanese sequence /dzi/, variation in the pronunciation has been observed: it can be realized as voiced affricate [dzi] or as voiced fricative [zi]. Traditionally, this variation was attributed to word-position: word-initial /dzi/ would be realized as the voiced affricate [dzi], and word-internal as the voiced fricative [zi] [17]. More recent studies, however, have found that the variation appears to be primarily linked to speech rate, with the voiced fricative [zi] occurring in quicker speech, and the voiced affricate [dzi] with a slower speech rate [18, 19]. For this reason, we transcribe this sequence in the following as [(d)zi].

In addition to testing a potential difference in performance, we include self-reported input of English by our participants. [20] demonstrated that self-assessment of English input could be used as a reliable tool to predict phonetic variation in the adaptation of English final consonants by Italian speakers. The present study tests whether the amount of self-reported received English input of Japanese listeners can similarly predict variation in the ability to perceive the contrast between [(d)zi] – [zi] and [tei] – [ti], with the expectation that a self-reported greater exposure to English positively corresponds to discrimination performance.

2. Perception experiment

For the present perception experiment, we employed an online AX discrimination task created with the software Experiment Designer (ED; [21]). The experiment was approved by the ethics committee of the University of Amsterdam. The task took approximately 10 minutes to complete and could be carried out in the comfort of the participant’s home.

2.1. Participants

39 Japanese native speakers (26 female, 12 male, 1 non-binary) participated in the online experiment, with ages ranging from 21 to 63 years at testing. During the selection process, we controlled for area of residence to minimize linguistic differences: all participants originated from and lived in the greater Tokyo region of Japan (including Tokyo, Kanagawa, Saitama, and Chiba). Furthermore, we controlled for linguistic environment to ensure that the participants’ perceptual input was not influenced substantially by another language; all participants grew up in monolingual Japanese households. Participants not fulfilling these criteria were excluded from the experiment (this applied to two, resulting in the given 39).

2.2. Speaker and recordings

Our speaker was a female native from the Tokyo region of Japan (Saitama prefecture), aged 27 at the time of recording. Due to her high level of English, this speaker could comfortably produce the sequence [zi], as attested perceptually and via inspection of the resulting spectrograms by the authors. The speaker produced several tokens of each stimulus, all recorded in the sound-attenuated booth of the Speech Lab at the University of Amsterdam, with a Neumann TLM103 cardioid condenser microphone at a 44.1 kHz sampling rate.

2.3. Stimuli

A total of 83 pairs of disyllabic Japanese non-words were created as stimuli for the online experiment. 40 pairs made up the experimental items: 20 tested the discrimination of [ti] – [tei], and 20 the discrimination of [zi] – [(d)zi]. Out of each of

these 20 experimental pairs, 10 pairs consisted of ‘same’ stimuli, i.e. two tokens of the same item, and 10 pairs consisted of ‘different’ stimuli. The latter were presented in one of two possible orders, i.e., in a fixed order for all participants. In Table 1, all experimental items (and for different stimuli also their order of presentation) are given.

For the 10 stimuli representing [(d)zi] in the different pairs, five were phonetically realized as [zi], and five as [dzi]. In the same pairs for [(d)zi], only [zi] was used, as our speaker produced [zi] more often than [dzi].

An illustration of a [ti] – [tei] experimental pair is given at the top of Figure 1. We can see that there is a clear difference in simple burst with some aspiration noise for [hiti] on the left and an affricate with longer friction noise following the burst for [hitei] on the right side. An example for a [zi] – [zi] experimental pair is given at the bottom of Figure 1. Here, the only difference between [ruzi] on the left and [ruzi] on the right is the starting frequency of the high-intensity friction noise of the sibilants, which is much higher for the alveolar [z] in the left spectrogram (its beginning is just visible at the upper edge, at approx. 8,500 Hz). Pairs with [zi] – [dzi] have the additional acoustic difference of a silence (the closure phase) before the alveolopalatal fricative for [dz], not illustrated here.

Further 40 pairs were fillers. Like the experimental pairs, the first syllable of filler pairs was always the same, whereas the second syllable could be the same or different. The phoneme combinations used to compile the filler items are all phonotactically licit in Japanese. Half of the filler pairs contained the same non-words, half different ones. Example for different filler pairs are: [keni] – [kegi], [bopi] – [bopa] and [miei] – [minu]. The remaining three pairs were catch trials that consisted of non-word pairs differing both in first and second syllable, e.g., [kega] – [teei]. These catch trials were deployed as a means to filter out participants who were not paying attention to the stimuli. If a participant marked a minimum of two out of three catch trials as ‘same’, thus incorrectly, their responses would be excluded from the results. As only four participants responded incorrectly to one out of three catch trials, no participant had to be excluded.

Stimuli pairs were created in Praat [22]. Same stimuli pairs always contained two different tokens of the same non-word. The inter-stimulus interval was always 800 ms. This ensured participants accessed their phonological knowledge instead of simply comparing acoustic similarity (cf. [23]).

Table 1: *Experimental items.*

[tei] – [ti]		[(d)zi] – [zi]	
same	different	same	different
[dotei] - [dotei]	[deti] - [detei]	[gazi] - [gazi]	[pizi] - [pizi]
[gitei] - [gitei]	[hiti] - [hitei]	[tezi] - [tezi]	[mizi] - [mizi]
[hetei] - [hetei]	[meti] - [metei]	[bozi] - [bozi]	[ruzi] - [ruzi]
[nete] - [nete]	[nutei] - [nutei]	[hezi] - [hezi]	[dazi] - [dadzi]
[pitei] - [pitei]	[weti] - [wetei]	[puzi] - [puzi]	[zezi] - [zedzi]
[retei] - [retei]	[beti] - [betei]	[juzi] - [juzi]	[wozi] - [wozi]
[tetei] - [tetei]	[hotei] - [hoti]	[wazi] - [wazi]	[buzi] - [buzi]
[watei] - [watei]	[rutei] - [ruiti]	[zuuzi] - [zuuzi]	[kedzi] - [kezi]
[zetei] - [zetei]	[witei] - [witi]	[zazi] - [zazi]	[gudzi] - [guzi]
[zutei] - [zutei]	[zatei] - [zati]	[zoz] - [zoz]	[nuuzi] - [nuuzi]

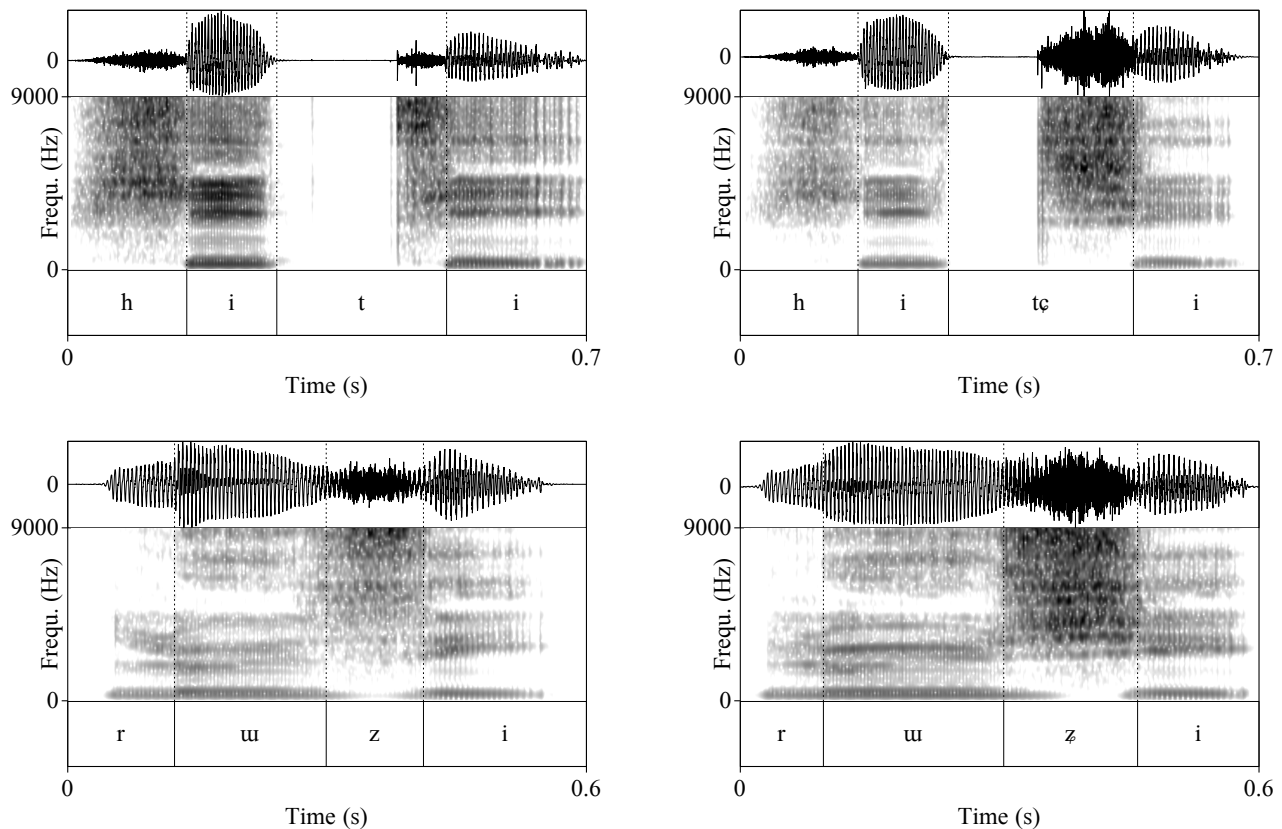


Figure 1: Oscillograms and spectrograms of the [ti] – [tei] pair [hiti] – [hitei] (top), and the [zi] – [zi] pair [ruzi] – [ruzi] (bottom).

2.4. Procedure

Participants were asked to complete the online AX discrimination task by listening to two Japanese bisyllabic non-words at a time, and indicating for each pair whether they perceived the two words to be the same or different. This was done by pressing the left or right arrow keys on their keyboard, where one key corresponded to ‘same’ and the other ‘different’. The participants were informed and instructed about the task through a short text in Japanese, where it was also explained that the amount of ‘same’ and ‘different’ items might not be balanced.

The experiment was divided into two trial blocks: the first containing 42 stimulus pairs and the second 41 stimulus pairs. The trial blocks were separated by a self-paced break. In the first block, participants were asked to select the left arrow key for ‘same’ and the right arrow key for ‘different’. After the break, the response keys were switched. In order to avoid confusion, the responses corresponding to arrow keys were listed at the bottom of the screen for the duration of the whole experiment. The order of appearance of stimulus pairs was pseudo-randomized such that experimental items were spaced apart by a minimum of two filler items. In addition, experimental items did not occur within the first three stimuli pairs in block 1. Catch trial pairs were presented in fixed positions as pairs 4, 50 and 79.

2.5. Self-report on English input

At the end of the experiment, participants had to answer three questions on the amount of English input they received:

- (1) How many hours a week are you exposed to the English language? (Through social media, movies and series, music, school, interactions with English native speakers)
- (2) How many hours a week do you speak English?
- (3) How many months did you spend in an English-speaking country?

For the first two questions, the response options were ranging from “0” to “15 or more hours”, for the last question the range was from “0” to “18 or more months”, all with steps of 1 in-between.

For each participant, we calculated an English input score by adding up the values they provided for each answer. This score could range from 0, corresponding to having had no English input at all, to 48, corresponding to having had very frequent input and having spent at least 18 months in an English-speaking country. The maximum our participants had was 37, see Figure 1 for the distribution of their English input scores.

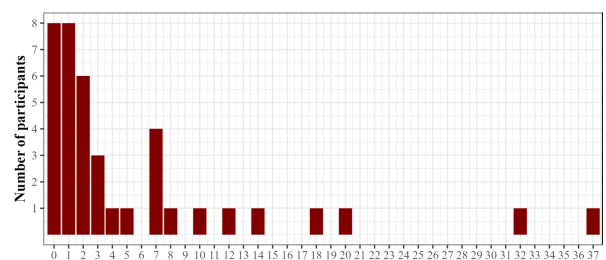


Figure 2: Distribution of English input scores (from 0 to the occurring maximum of 37) by participants.

2.6. Statistical analysis

The statistical analysis was carried out with R [24] in RStudio using *lm4* [25] and *lmerTest* [26] packages. To investigate whether participants were better at discriminating [tei] – [ti] than [(d)zi] – [zi], we looked only at different pairs of the experimental items. A generalized linear mixed-effects model was fitted with *Correctness* of the responses to the experimental items (binary; with 1 for correct and 0 for incorrect) as a function of the categorical predictor *Contrast* (categorical: [ti] – [tei] or [zi] – [(d)zi]; coded with a sum-to-zero orthogonal contrast). Random intercepts were fitted for *Participant* and *Item*.

For the question on whether self-reported English input had an effect on perception of the two contrasts, a generalized linear mixed-effects model was fitted with *Correctness* as a function of the interaction between the continuous predictor *English score* and the categorical predictor *SameDifferent* (coded with a sum-to-zero orthogonal contrast). Random slopes were fitted for the interaction between *SameDifferent* and *Participant*, and for the interaction between *English score* and *Item*.

Additionally, descriptive statistics were drawn up by calculating the number and percentages of correct responses per participant.

3. Results

Participants' performance on 'different' experimental items for the [(d)zi] – [zi] contrast was 90% on average, ranging from 40% correct (one participant) to 100% correct (18 participants). For the [tei] – [ti] contrast, participants performed with an average accuracy of 98%, ranging from 90% (7 participants) to 100% (32 participants), see Fig. 3. The statistical analysis showed that participants' odds of responding accurately on 'different' items of [tei] – [ti] were 7.28 times higher than the odds of responding accurately to 'different' items of [(d)zi] – [zi], and this main effect of *Contrast* on *Accuracy* was highly significant ($p < 0.001$, 95% CI = 2.329 .. 22.778).

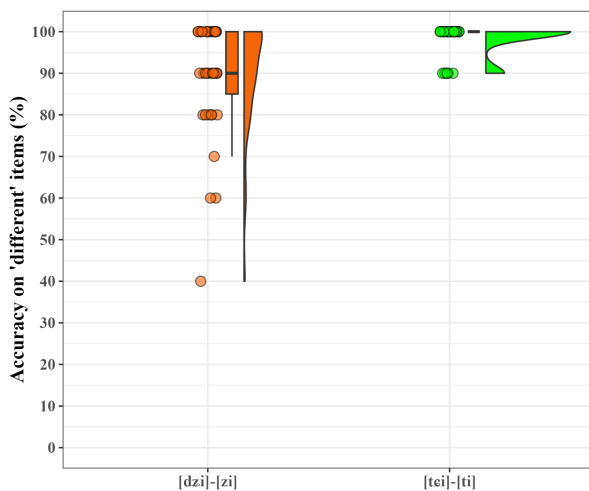


Figure 3: Accuracy for “different” items (in %) for [(d)zi] – [zi] (including a foreign sequence) on the left and [tei] – [ti] (including a loan sequence) on the right.

The statistical analysis for the influence of English input score showed that there was no significant main effect of *English score* ($p = 0.411$, 95% CI = 0.968 .. 1.081) and no significant

interaction with *SameDifferent* ($p = 0.636$; 95% CI = 0.891 .. 1.073).

4. Discussion

On the basis of the results by [15] we expected that Japanese participants would generally perform well on the [(d)zi] – [zi] contrast, which our current results confirmed.

We inquired into the discrimination of the [(d)zi] – [zi] contrast (containing a non-native sequence) in comparison to that of the [tei] – [ti] contrast (containing two native sequences, one only occurring in loanwords). Our results show that there was a significant effect of contrast on the participants' accuracy on 'different' experimental items, meaning that they performed better on the [tei] – [ti] contrast than on the [(d)zi] – [zi] contrast. This is an important finding, because not only does it provide evidence for the observation that [ti] has been integrated into the Japanese phonotactic system and participants are better at distinguishing this than the non-integrated [zi], it also shows that our participants generally performed very well on the [(d)zi] – [zi] contrast, with 29 out of 39 participants scoring 90% or higher, though with a high amount of interspeaker variability. Considering how well our participants performed on the illegal sequence, it leads us to predict that loanwords with a faithful adaptation of [zi] are likely to appear in Japanese in the near future.

The second hypothesis, that the degree of self-reported English input correlates with the performance in the discrimination, was not supported. As the interaction of accuracy and English input was not significant, we cannot confirm on the basis of our results that participants with higher English input scores are more sensitive to both contrasts. This result could be attributed to a number of explanations, the first one being that English input alone is not sufficient to account for Japanese speakers' ability to perceive these contrasts, and that other variables, such as level of education or age, need to be included. Second, the system of scoring English input applied in the present study may have been oversimplified, in the sense that having spent two months in an English speaking country might have a different effect on a speakers' performance than a weekly input of two hours.

Taking an English proficiency test (as traditionally done in L2 experiments, but not in experiments on loan adaptations) is a way to provide a more reliable judgment of participants' knowledge in English, though this is of course also far more time consuming.

5. Conclusions

Our results provided evidence for the hypothesis that Japanese listeners are perceptually better at distinguishing the loan sequence [ti] than the foreign sequence [zi] from similar native sequences. Their performance for the latter is quite advanced, which leads us to the conclusion that perception is not only guided by existing native sequences, but also by illegal phonotactic sequences the speakers know from an L2. We therefore think it likely that the sequence [zi] is likely to enter the Japanese language via English loanwords in the near future.

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7. References

- [1] E.D. Polivanov, “La perception des sons d’une langue étrangère.” *Travaux du Cercle Linguistique de Prague*, vol. 4, pp. 79–96, 1931. [English translation: The subjective nature of the perceptions of language sounds. In E.D. Polivanov. *Selected works: articles on general linguistics*. The Hague: Mouton, 1974, pp. 223–237]
- [2] E. Dupoux, K. Kakehi, Y. Hirose, C. Pallier and J. Mehler, “Epenthetic vowels in Japanese: a perceptual illusion.” *Journal of Experimental Psychology: Human Perception and Performance* vol. 25, pp. 1568–1578, 1999.
- [3] B. Kabak and W. Idsardi, “Perceptual distortions in the adaptation of English consonant clusters: Syllable structure or consonantal contact constraints?” *Language and Speech*, vol. 50, no. 1, pp. 23–52, 2007.
- [4] K. Durvasula, and J. Kahng, “Illusory vowels in perceptual epenthesis: The role of phonological alternations.” *Phonology*, vol. 32, pp. 385–416, 2015.
- [5] Y. Wang, R. L. Bundgaard-Nielsen, B. J. Baker and O. Maxwell, “Native phonotactic interference in L2 vowel processing: Mouse-tracking reveals cognitive conflicts during identification.” *Proceedings of Interspeech 2022*. Incheon, Korea, Sep. 2022, pp. 5223–5227.
- [6] E. Haugen, “The analysis of linguistic borrowings.” *Language* Vol. 26, pp. 210–231, 1950.
- [7] C. Paradis and D. LaCharité, “Preservation and minimality in loanword adaptation.” *Journal of Linguistics* vol. 33, pp. 379–430, 1997.
- [8] S. Hamann and D. Li, “Adaptation of English onset clusters across time in Hong Kong Cantonese: the role of the perception grammar.” *Linguistics in Amsterdam* vol. 9, pp. 56–76, 2016.
- [9] H. Kubozono, “Loanword phonology,” In H. Kubozono (ed.) *Handbook of Japanese Phonetics and Phonology*. Berlin: De Gruyter Mouton, pp. 313–362, 2015.
- [10] G. Kay, “English loanwords in Japanese,” *World Englishes*, vol. 14, no. 1, pp. 67–76, 1995.
- [11] G. Pintér, “The emergence of new consonant contrasts.” In H. Kubozono (ed.) *Handbook of Japanese Phonetics and Phonology*. Berlin: De Gruyter Mouton, pp. 121–166, 2015.
- [12] C. Crawford, “An evolutionary account of loanword-induced sound change in Japanese,” *University of Pennsylvania Working Papers in Linguistics*, vol. 14, no. 1, pp. 55–70, 2008.
- [13] C. Crawford, *Adaptation and transmission in Japanese loanword phonology*. PhD thesis, Cornell University, 2009.
- [14] M. Irwin, *Loanwords in Japanese*. Amsterdam: John Benjamins, 2011.
- [15] F. E. Daulton, “Appendix 2: The List of Common Loanwords Corresponding to the BNC 3000.” In *Japan’s Built-in Lexicon of English-based Loanwords*. Clevedon: Multilingual Matters, pp. 133–155, 2008.
- [16] A. Aličehajić and S. Hamann, “The discrimination of [zi]-[dzi] by Japanese listeners and the prospective phonologization of /zi/.” *Proceedings of Interspeech*, Incheon, Korea, Sep. 2022, pp. 1322–1326.
- [17] Y. Amanuma, K. Otsubo and O. Mizutani, *Nihongo onseigaku* [Japanese phonetics]. Tokyo: Kuroshio Shuppan, 1983.
- [18] K. Maekawa, “Coarticulatory reinterpretation of allophonic variation: Corpus-based analysis of /z/ in spontaneous Japanese.” *Journal of Phonetics*, vol. 38, no. 3, pp. 360–374, 2010.
- [19] S. Matsui, K. Iwamoto and R. Mazuka, “Development of allophonic realization until adolescence: A production study of the affricate-fricative variation of /z/ among Japanese children.” *Proceedings of Interspeech*, Incheon, Korea, Sep. 2022, pp. 739–743.
- [20] V. Miatto, S. Hamann and P. Boersma, “Self-reported L2 input predicts phonetic variation in the adaptation of English final consonants into Italian,” *Proceedings of the 19th ICPHS*, 2019, pp. 949–953.
- [21] D.J. Vet, ED - Experiment Designer, [Computer program], <https://www.fon.hum.uva.nl/dirk/ed.php>, 2023.
- [22] P. Boersma and D. Weenink, *Praat: doing phonetics by computer* [Computer program], 2022, version 6.2.05, <http://www.praat.org/>
- [23] G. M. Miglietta, and A. Calabrese, “Conditioned allophony in speech perception: An ERP study,” *Brain and Language*, vol. 126, no. 3, pp. 285–290, 2013.
- [24] R Development Core Team. *R: a language and environment for statistical computing*. 2020, version 4.3.3. www.r-project.org
- [25] D. Bates, M. Mächler, B. Bolker and S. Walker, “Fitting linear mixed-effects models using lme4,” *Journal of Statistical Software*, 2015. <http://arxiv.org/abs/1406.5823>
- [26] A. Kuznetsova, P. B. Brockhoff & R. H. B. Christensen, “lmerTest Package: Tests in Linear Mixed Effects Models,” *Journal of Statistical Software*, 2017, vol. 82, pp. 1–26, <https://doi.org/10.18637/jss.v082.i13>